

PENHAS DOURADAS, Portugal

WMO: 085680

Lat: 40.4114N

Lon: 7.5586W

Elev: 4531

StdP: 12.44

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	25.6	-3.9	5.3	44.3	3.2	7.7	44.3	41.6	36.0	38.3	35.7	10.9	130	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.0	81.5	59.8	78.6	58.7	76.0	58.1	62.9	75.1	61.5	73.7	60.1	72.3	8.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.5	86.7	67.5	56.9	81.7	65.9	55.5	77.6	64.6	30.8	75.2	29.7	73.8	28.7	72.4	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
35.6	30.4	26.1	DB	19.3	86.2	3.7	2.2	16.6	87.8	14.5	89.1	12.4	90.4	9.8	92.0
			WB	17.4	65.8	3.8	2.0	14.7	67.3	12.4	68.5	10.3	69.6	7.6	71.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	38.1	38.9	42.2	44.4	50.8	58.8	64.0	65.0	59.5	51.5	42.2	39.9
	DBStd	11.96	6.28	6.58	7.17	7.74	8.29	8.25	7.59	7.09	6.67	6.98	6.03	5.73
	HDD50	1899	371	313	259	206	96	20	2	1	6	65	243	317
	HDD65	5825	835	731	708	617	446	215	112	89	186	422	685	779
	CDD50	1779	1	2	16	39	119	285	437	467	292	110	8	3
	CDD65	230	0	0	0	0	4	30	82	90	22	2	0	0
	CDH74	1028	0	0	0	0	9	100	394	461	61	3	0	0
	CDH80	146	0	0	0	0	0	5	55	80	6	0	0	0
(14) Wind	WSAvg	10.8	11.2	11.3	11.8	11.9	10.1	10.3	11.1	9.8	8.7	10.3	11.6	11.5
(15) Precipitation	PrecAvg	65.7	8.9	7.9	5.3	5.6	4.8	2.8	0.9	0.7	2.7	7.2	9.5	9.7
	PrecMax	105.1	21.3	21.3	20.3	17.4	9.3	10.0	3.8	3.0	9.8	18.5	27.0	31.6
	PrecMin	40.3	0.0	0.8	0.0	0.9	0.5	0.2	0.0	0.0	0.1	0.2	0.2	1.2
	PrecStd	16.6	5.7	6.0	4.6	3.6	2.8	2.7	1.0	0.7	2.1	5.0	6.9	7.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	57.3	63.4	67.4	75.2	80.6	84.4	86.6	79.2	73.5	60.0	58.0
		MCWB	42.2	43.0	47.3	52.6	57.2	59.5	60.6	60.6	57.7	53.7	48.4	43.6
	2%	DB	52.2	54.0	59.1	63.9	70.6	76.8	81.6	82.1	75.4	68.3	56.0	53.5
		MCWB	40.4	42.8	45.5	50.9	55.0	58.5	60.2	59.7	56.8	52.2	46.2	41.8
	5%	DB	49.0	50.9	55.8	60.2	67.1	74.2	78.5	79.0	72.7	64.3	52.8	50.7
		MCWB	40.0	40.9	44.2	49.1	53.8	57.7	59.3	58.6	55.6	51.1	45.4	41.2
	10%	DB	46.4	47.8	52.6	56.7	63.7	71.1	75.5	76.1	69.7	61.0	50.4	47.8
		MCWB	40.3	40.0	42.6	47.2	52.1	56.9	58.3	58.0	54.8	50.2	44.9	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	48.1	50.2	55.3	60.7	63.7	65.1	64.1	61.3	58.5	53.6	48.7
		MCDB	50.2	51.5	58.6	64.9	70.0	73.7	77.8	78.0	71.8	64.5	56.1	51.0
	2%	WB	45.8	45.7	48.0	52.2	57.8	61.8	63.1	62.1	59.3	56.3	50.7	46.9
		MCDB	47.2	49.4	54.2	61.2	66.7	72.3	75.7	75.7	70.2	61.6	52.5	48.7
	5%	WB	44.1	43.9	46.3	50.1	55.3	60.1	61.5	60.9	57.9	54.5	48.7	45.2
		MCDB	45.6	47.6	52.1	58.6	64.7	70.6	74.0	74.4	68.4	59.5	50.4	47.3
	10%	WB	42.4	42.0	44.5	48.0	53.1	58.3	60.0	59.6	56.6	52.9	46.8	43.5
		MCDB	44.5	45.6	50.2	55.6	61.9	68.9	72.5	73.0	66.2	57.8	48.8	45.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.8	11.9	12.5	13.3	14.4	15.9	16.0	14.0	10.7	9.2	9.6
		MCDBR	12.8	13.8	15.6	16.1	16.4	16.4	17.0	17.5	16.5	14.2	12.1	12.4
		MCWBR	9.8	10.3	11.0	10.6	11.2	12.0	11.3	10.1	9.7	10.1	10.2	10.2
	5% WB	MCDBR	9.9	11.8	13.7	15.3	15.4	15.9	16.0	16.4	14.9	11.4	9.4	10.0
		MCWBR	9.2	10.2	10.8	10.9	11.1	12.6	12.1	11.2	10.0	9.3	9.0	9.6
(40) Clear-Sky Solar Irradiance	taub		0.265	0.276	0.301	0.304	0.310	0.314	0.304	0.315	0.306	0.297	0.275	0.264
	taud		2.602	2.574	2.515	2.510	2.512	2.515	2.522	2.480	2.537	2.585	2.603	2.602
	Ebn at Noon		292	303	304	308	306	304	306	300	298	291	286	284
	Edh at Noon		21	25	30	32	33	33	33	33	29	25	21	20
(44) All-Sky Solar Radiation	RadAvg		638	950	1286	1629	2018	2313	2441	2161	1643	1056	698	571
	RadStd		99	137	189	206	219	138	80	99	126	124	123	85

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92

Nomenclature: See separate page